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Renesas VoLTE Power Consumption

Report of Evidence of Use

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Patent Number: US8310969B2

1. A mobile device used in a wireless network system, comprising: a modem processor electrically connected to said wireless network system to receive/transmit a network packet, said modem processor including an IP packet protocol module and a storage buffer; and a data processor connected via said modem processor to said wireless network system for network service; wherein when said data processor is in a hibernation mode, said IP packet protocol module unpacks said received network packet to access a packet data and said modem processor selectively wakes up said data processor according to said packet data; wherein when said modem processor determines that said packet data needs to be processed by said data processor, then said modem processor wakes up said data process by sending said packet data to said data processor; wherein when said modem processor determines that said packet data is not necessary to wake up said data processor, then said modem processor sends said packet data to said storage buffer.

Element 1:

1. A mobile device used in a wireless network system, comprising:

Supporting Evidence:

CLAIM 8. *Reference to Claim Preamble:*

DETAILED TECHNOLOGY MAPPING

A mobile device used in a wireless network system, comprising:

Optimized IMS Architecture to be used in a phone platform and various products.

4. ARCHITECTURE

This whitepaper discusses the IMS architecture for the voice and video use cases. It presents an optimized architecture for VoLTE audio processing for audio and video calls that also provides an upgrade path to video processing in video calls and allows power saving when only a voice call is active or the video call is downgraded to audio only.

In a phone platform, the change from circuit switched voice support to packet switched and circuit switched voice, i.e. VoLTE, requires changes in the software architecture to minimize the power consumption. The packet voice introduces an IMS telephony application and the IMS stack to the application processor. Additionally, it also introduces an IP stack and packet voice client to handle the RTP/RTCP protocol and jitter buffer to the modem.

The power consumption of video & display hardware is potentially so high that having the AP in sleep mode would not make big difference to the power consumption, i.e when video is active, the audio processing may be done either in the AP or the modem. The location of the audio processing with active video is more based on implementation simplicity as having active video nullifies any audio power saving efforts. The Renesas Mobile architecture, implementable on a variety of products, allows flexible partitioning of OEM video telephony for cellular and WIFI connectivity.

In the case that the IMS call also has a video component, the video RTP/RTCP is routed to the AP. Routing video to the modem would not result in power consumption savings as the AP needs anyhow to be active during the video call. Renesas Mobile's VoLTE architecture provides a more efficient voice service than the 3G voice as Renesas Mobile supports the optimization of audio routing.

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 2:

a modem processor electrically connected to said wireless network system to receive/transmit a network packet,

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

a modem processor electrically connected to said wireless network system to receive/transmit a network packet,

Modem is used to transmit/receive LTE signals.

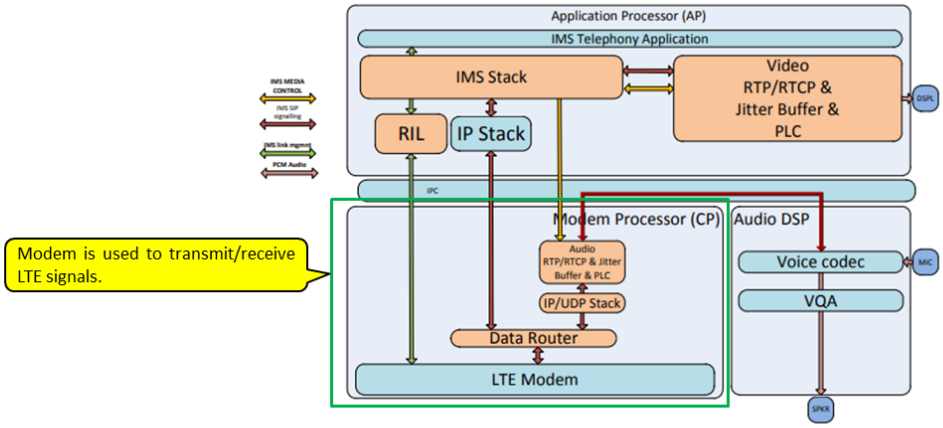


Figure 1-2 Optimized architecture for voice

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 3:

said modem processor including an IP packet protocol module and

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

said modem processor including an IP packet protocol module and

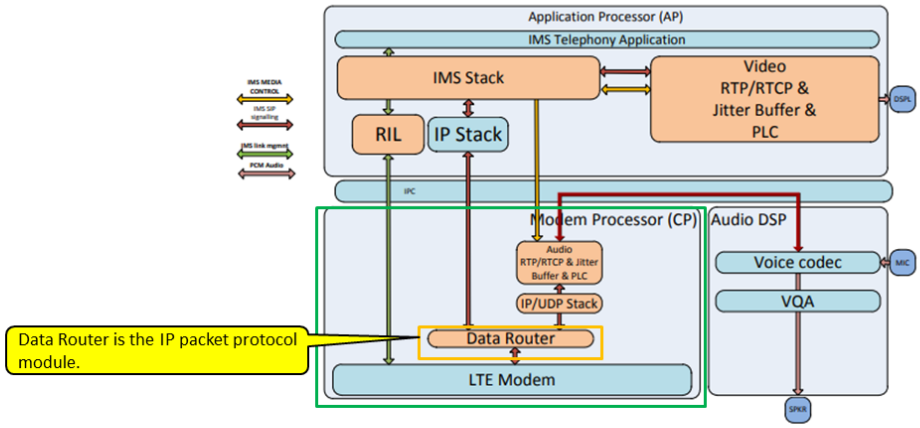


Figure 1-2 Optimized architecture for voice

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 4:

a storage buffer; and

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

a storage buffer; and

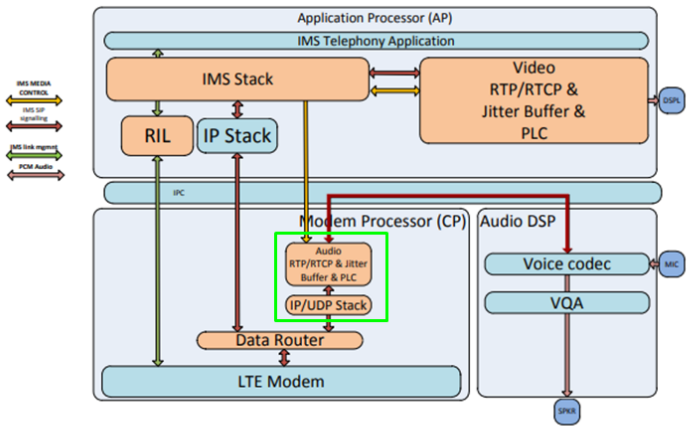


Figure 1-2 Optimized architecture for voice

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Element 5:

a data processor

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

a data processor

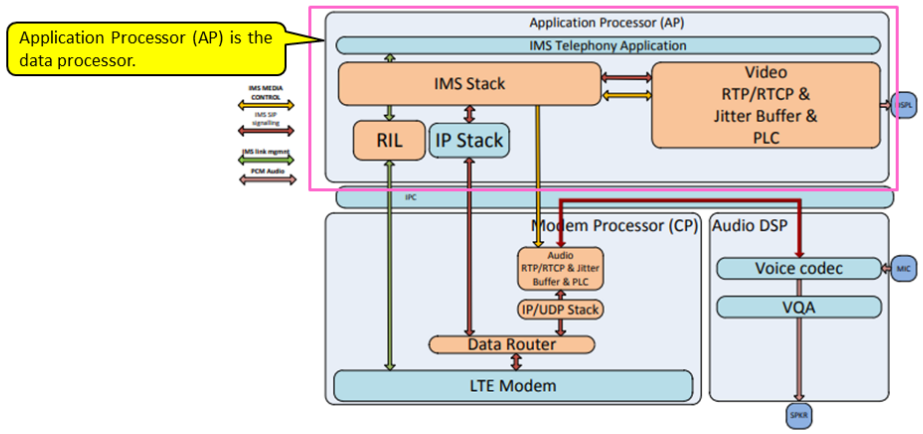


Figure 1-2 Optimized architecture for voice

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 6:

connected via said modem processor to said wireless network system for network service;

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

connected via said modem processor to said wireless network system for network service;

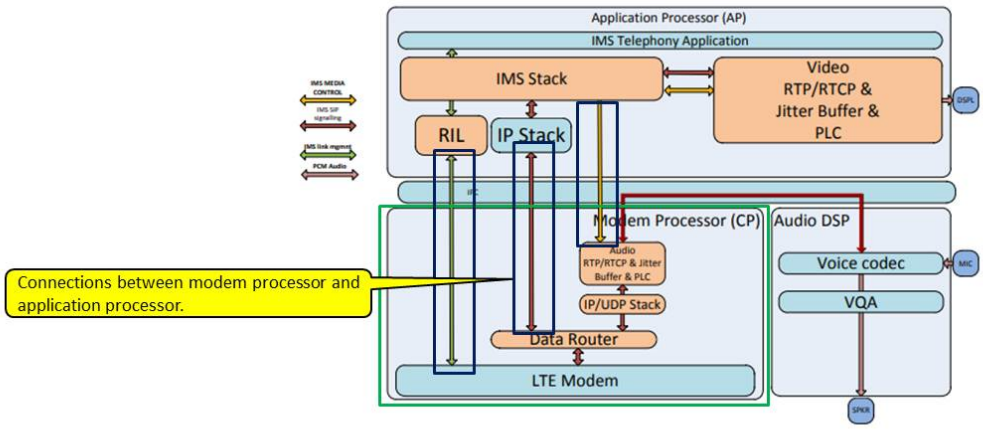


Figure 1-2 Optimized architecture for voice

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 7:

wherein when said data processor is in a hibernation mode, said IP packet protocol module unpacks said received network packet

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

wherein when said data processor is in a hibernation mode, said IP packet protocol module unpacks said received network packet

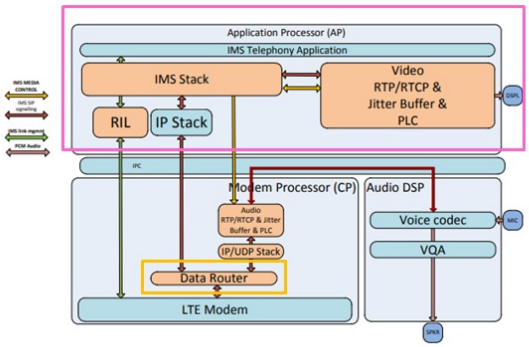


Figure 1-2 Optimized architecture for voice

Some other public papers on LTE Telephony and battery life discuss the partitioning options for the IMS Protocol stack and audio processing onto the handset processors. Renesas Mobile thinks that the application processor is a straightforward solution with the least complexity to process the VoLTE audio e.g. with Linux based operating systems and application related restrictions. But we also recognize that there are other more complex solutions which allow for higher potential power saving. The modem processor provides better power saving opportunities for audio, and in most platform architectures the legacy voice call audio handling is done in the modem processor. Thus in practice, when the VoLTE user plane audio handling is implemented in the modem processor, the application processor can enter sleep mode.

As seen in Figure 1-2, in an optimized architecture the voice processing is implemented in the modem processor. The architecture split is made so that the processing capability and power consumption are optimised. The VoLTE audio stream is now routed to the IP stack of the Modem processor. In the case that there are no active IP connections to applications in the AP, it may enter sleep mode during a VoLTE call, enabling lower power consumption.

Data Router unpacks the received packets and routing decisions are made according to the type of packets (Audio/Video).

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 8:

to access a packet data and

Supporting Evidence:

CLAIM 8. *Reference to Claim Element: Cont (..)*

DETAILED TECHNOLOGY MAPPING

to access a packet data and

LTE is designed as a dedicated packet based system and does not have Circuit Switched (CS) channels which are most suitable for speech scenarios like VoLTE. When CS voice call types of scenarios are considered, the LTE system suffers from the fact that IP packet data traffic exposes the system to a completely different behavior. IP packet data scenarios exploit a more dynamic burst type of behavior where burst sizes and intervals may vary significantly, whereas in CS speech the traffic is much more predictable with respect to the burst size and interval.

In the case that the IMS call also has a video component, the video RTP/RTCP is routed to the AP. Routing video to the modem would not result in power consumption savings as the AP needs anyhow to be active during the video call. Renesas Mobile's VoLTE architecture provides a more efficient voice service than the 3G voice as Renesas Mobile supports the optimization of audio routing.

In LTE system, Audio/Video Signals are received in the form of packets.

As seen in Figure 1-2, in an optimized architecture the voice processing is implemented in the modem processor. The architecture split is made so that the processing capability and power consumption are optimised. The VoLTE audio stream is now routed to the IP stack of the Modem processor. In the case that there are no active IP connections to applications in the AP, it may enter sleep mode during a VoLTE call, enabling lower power consumption.

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

Element 9:

said modem processor selectively wakes up said data processor according to said packet data; wherein when said modem processor determines that said packet data needs to be processed by said data

processor, then said modem processor wakes up said data process

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

said modem processor selectively wakes up said data processor according to said packet data; wherein when said modem processor determines that said packet data needs to be processed by said data processor, then said modem processor wakes up said data processor;

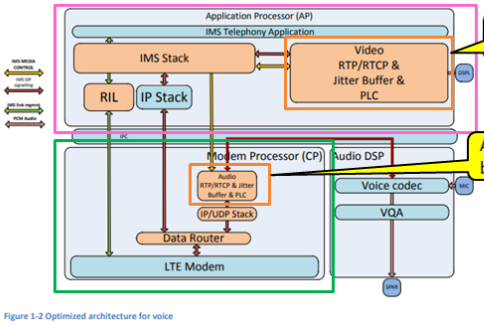


Figure 1-2 Optimized architecture for voice

Video content routed to application processor for processing.

Audio content processed by modem processor itself.

Application processor remains in sleep mode until modem processor sends packets to wake up the application processor.

Packets related to the video part are routed to the application processor by modem processor for processing whereas packets related to the audio part are processed by modem processor itself.

In the case that the IMS call also has a video component, the video RTP/RTCP is routed to the AP. Routing video to the modem would not result in power consumption savings as the AP needs anyhow to be active during the video call. Renesas Mobile's VoLTE architecture provides a more efficient voice service than the 3G voice as Renesas Mobile supports the optimization of audio routing.

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Element 10:

by sending said packet data to said data processor;

Supporting Evidence:

CLAIM 8. *Reference to Claim Element:*

DETAILED TECHNOLOGY MAPPING

by sending said packet data to said data processor;

In the case that the IMS call also has a video component, the video RTP/RTCP is routed to the AP. Routing video to the modem would not result in power consumption savings as the AP needs anyhow to be active during the video call. Renesas Mobile's VoLTE architecture provides a more efficient voice service than the 3G voice as Renesas Mobile supports the optimization of audio routing.

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Element 11:

wherein when said modem processor determines that said packet data is

not necessary to wake up said data processor, then said modem processor sends said packet data to said storage buffer.

Supporting Evidence:

DETAILED TECHNOLOGY MAPPING

CLAIM 8. *Reference to Claim Element:*

wherein when said modem processor determines that said packet data is not necessary to wake up said data processor, then said modem processor sends said packet data to said storage buffer.

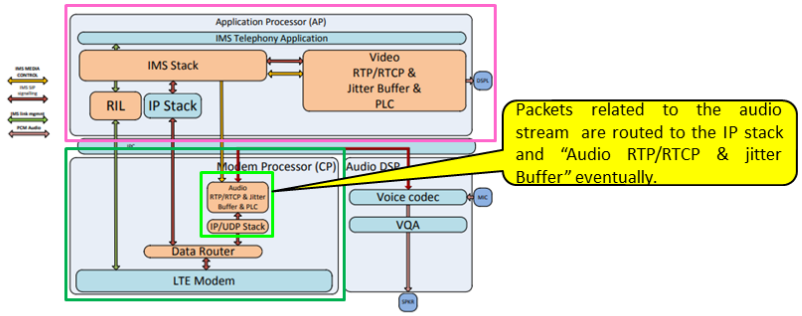


Figure 1-2 Optimized architecture for voice

As seen in Figure 1-2, in an optimized architecture the voice processing is implemented in the modem processor. The architecture split is made so that the processing capability and power consumption are optimised. The VoLTE audio stream is now routed to the IP stack of the Modem processor. In the case that there are no active IP connections to applications in the AP, it may enter sleep mode during a VoLTE call, enabling lower power consumption.

Application processor remains in sleep mode until video stream is received and routed to it by data router (part of modem processor).

SOURCE: https://ytd2525.files.wordpress.com/2013/02/renesas-mobile-volte-power-consumption-power-whitepaper-2013-02.pdf

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End of Report

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Our hourly rate is: $

Non-Disclosure Agreement:

We understand that for the purpose of assisting you in collecting data, you may disclose to us certain confidential information that you want to keep confidential; therefore, for good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, we agree as follows:

1. For a period of twelve (36) months from the date we receive your confidential information ("Effective Date"), we will not use or disclose the information, except: (a) in the event that such information (i) comes into the public domain not through us; (ii) is already known to us at the time of disclosure; (iii) is lawfully otherwise learnt or independently developed by us; or (iv) is required to be disclosed by law.

2. Since a breach of our obligations hereunder may result in irreparable harm to you, we agree that upon such breach or threat thereof, you shall be entitled to specific performance and other injunctive relief without posting a bond, and in the event of disclosures required by court order, we shall promptly notify you of the proceeding.

3. This agreement supersedes all prior discussions and writings and constitutes the entire agreement between you and us with respect to the subject matter hereof. This Agreement shall be binding upon the successors and assigns of the respective parties, and may not be modified except by an instrument executed by both parties.

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